

31 Oct 25

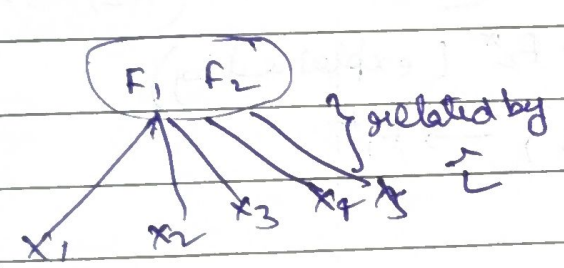
Sample Correlation Matrix

$$R = \begin{pmatrix} 1 & 0.2 & 0.96 & 0.42 & 0.01 \\ & 1 & 0.13 & 0.71 & -0.85 \\ & & 1 & 0.50 & 0.11 \\ & & & 1 & 0.79 \\ & & & & 1 \end{pmatrix}$$

Symm

$p = 5$
 $m = 2 \quad f_1, f_2$

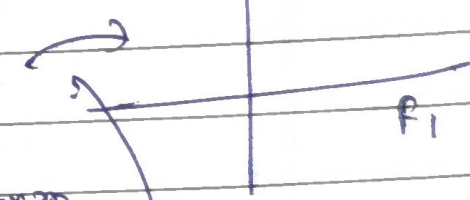
A:- $\hat{L} = [\sqrt{\lambda_1} e_1 \quad \sqrt{\lambda_2} e_2]$



$$= \begin{bmatrix} 0.56 & 0.82 \\ 0.78 & -0.52 \\ 0.65 & 0.75 \\ 0.94 & -0.105 \\ 0.80 & -0.51 \end{bmatrix}$$

5x2

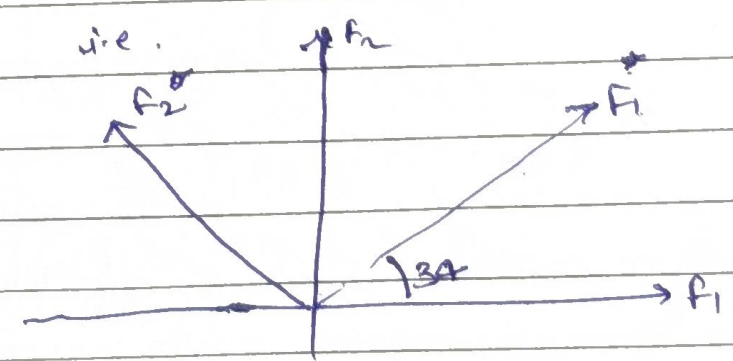
f_2
 $(0.56, 0.82)$



This plot does not give much meaning to us

So we try to rotate this by some angle & then plot it.

Here $\alpha = 34^\circ$



$$F^* = T^t F$$

$$\hat{L}^* = \hat{L} T^{m \times m}$$

5x2 5x2 2x2

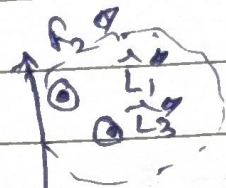
Rotational matrix

$$T = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}, \theta = 34^\circ$$

→ std. 2D Rotation Matrix

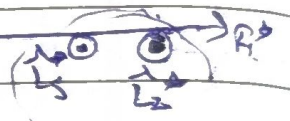
$$= \begin{pmatrix} 0.829 & 0.559 \\ -0.559 & 0.829 \end{pmatrix}$$

$$\hat{L}^* = \begin{pmatrix} F_1^* & F_2^* \\ 0.102 & 0.99 \\ 0.94 & -0.01 \\ 0.113 & 0.98 \\ 0.84 & 0.43 \\ 0.97 & -0.01 \end{pmatrix}$$



x_1, x_3 in same group

This is close to x_2, x_3 but not sure



(x_2, x_5) in same group

- $\{x_1, x_3\} \rightarrow F_2^*$ (explained by)
- $\{x_2, x_4, x_5\} \rightarrow F_1^*$

$$\underline{X} = \underline{\mu} + \underline{L}\underline{F} + \underline{\epsilon}$$

Linear Model i.e. idea of linear

dependency. But this fails to capture non-linear dependencies.

Hypothesis Testing :-

classmate

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Estimating the factor model by MLE Process.

$$\Sigma \leftrightarrow L, \Psi$$

Unknown

Sampling

(n random samples)

$$\sim N_p(\mu, \Sigma)$$

$$\begin{matrix} S \text{ or } R \\ \hat{L} \text{ or } \hat{\Psi} \end{matrix}$$

$$\hat{\Psi} = \text{diag}(S - \hat{L}\hat{L}^T)$$

Mostly focus on this

We are using likelihood function, so working in exponents has to be related to Normal.

Claims:

$\hat{L}$ is an estimator of L

$\hat{\Psi}$ is an estimator of Ψ

$$S_n = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(x_i - \bar{x})^T \rightarrow \text{MLE}$$

$$S = \frac{n S_n}{n-1}$$

Sample Variance

Unbiased Estimator

Null Hypothesis.

$$H_0 := \Sigma = LL^T + \Psi$$

Alternate

$H_1 := \Sigma$ is any arbitrary symmetric Positive definite p x p matrix.

Reval :-

$$l(\Sigma) = \log L = -\frac{n}{2} (\log |\Sigma| + \text{trace}(S_n^{-1} \Sigma)) + \text{const}$$

$$\text{also, } \hat{\mu}_{MLE} = \bar{x}$$

because of 2π factor
(Ignore the constant for maximisation)

$$l(\hat{\Sigma}_{H_1}) = -\frac{n}{2} (\log |\hat{\Sigma}_{H_1}| + p)$$

$\hat{\Sigma}_{H_1} \rightarrow$ MLE of Σ when H_1 is assumed to be holding.

$$\hat{\Sigma}_{H_1} = S_n \quad (\text{recall from past notes})$$

$$\text{trace}(S_n^{-1} \hat{\Sigma}_{H_1}) = \text{trace}(I_p) = p$$

Under H_0

$$l(\hat{\Sigma}_{H_0}) = \hat{L}\hat{L}^T + \hat{\Psi}$$

(using sample & assuming $H_0: \Sigma = LL^T + \Psi$)

$$= -\frac{n}{2} \left(\log |\hat{L}\hat{L}^T + \hat{\Psi}| + \text{trace}(S_n^{-1}(\hat{L}\hat{L}^T + \hat{\Psi})) \right)$$

Wilk's Lambda

$$\Lambda = \sup L(\Sigma, \mu)$$

$$\frac{\sup_{\Sigma, \mu \in \Theta_{H_0}} L(\Sigma, \mu)}{\sup_{\Sigma, \mu \in \Theta} L(\Sigma, \mu)} = \frac{L(\hat{\Sigma}_{H_0}, \bar{x})}{L(\hat{\Sigma}_{H_1}, \bar{x})}$$

$\Sigma, \mu \in \Theta$

all symm pd matrices (H_1)

$$\Sigma = LL^T + \Psi$$

factorisable

$$\Rightarrow -2 \ln \Lambda = -2 \left(l(\hat{\Sigma}_{H_0}) - l(\hat{\Sigma}_{H_1}) \right)$$

Type

$$\Rightarrow -2 \ln \Lambda = -2 \left[-\frac{n}{2} (\log |\hat{L}\hat{L}^T + \hat{\Psi}| + \text{trace}(S_n^{-1}(\hat{L}\hat{L}^T + \hat{\Psi}))) + \frac{n}{2} (\log |S_n| + p) \right]$$

⇒

$$-2 \ln \Lambda = n \log |\hat{L} \hat{L}^T + \hat{\Psi}| + n \text{trace} (S_n^{-1} (\hat{L} \hat{L}^T + \hat{\Psi}))$$

$$= n \log |S_n| - np.$$

Recall

$$-2 \ln \Lambda \xrightarrow[n \rightarrow \infty]{d} \chi^2_{(\mu)}$$

$$\mu = \dim \Theta - \dim \Theta_0$$

$$= \frac{p(p+1)}{2} - \left[pm + p - \frac{m(m-1)}{2} \right]$$

$$H_0: L L^T + \Psi = \Sigma$$

$$L: pm$$

$$\Psi: p$$

$$(pm + p) - \frac{m(m-1)}{2}$$

$$L^T \Psi^{-1} L = \Delta$$

↳ Diagonal matrix

This is to ensure constraint, +
to take care of rotation)

All the rotational transformation form
a group of order $\frac{m(m-1)}{2}$ where

m
↓
Dimension
of Rotation
matrix i.e.
($m \times m$)

(or no. of factors
in factor model)

~~is~~ ~~is~~ ~~is~~ is number of free
variable required to get the all possible
transformations

For sample .

Compute $-2 \ln \Lambda$

Check

$$-2 \ln \Lambda > \chi^2_{\mu, \alpha} \rightarrow \text{significance level}$$

⇓
reject H_0 .